



# Voicing and Airstream

Dr. Josef Fruehwald

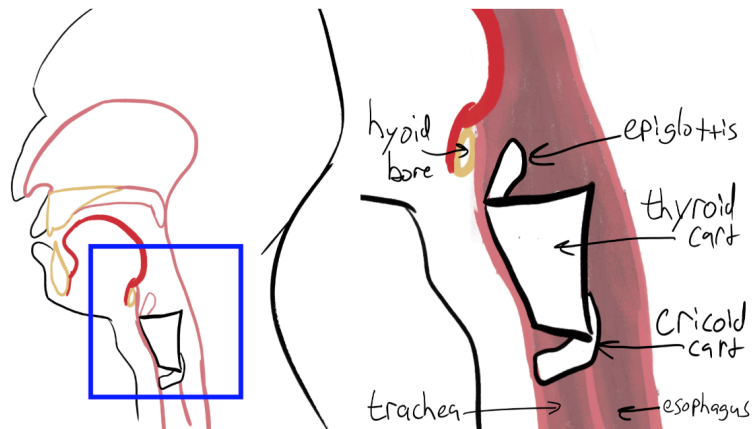
2026-09-08

## Voicing

Voicing

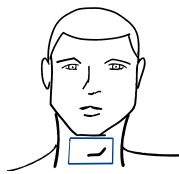
Previously:

## The Larynx

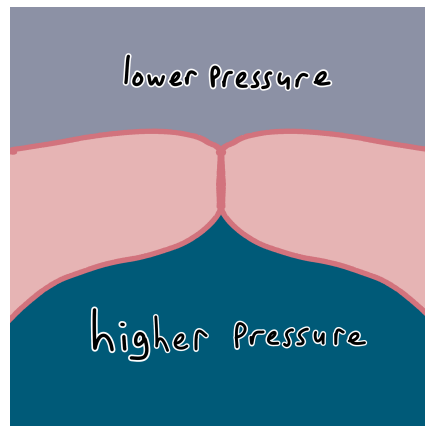


Voicing

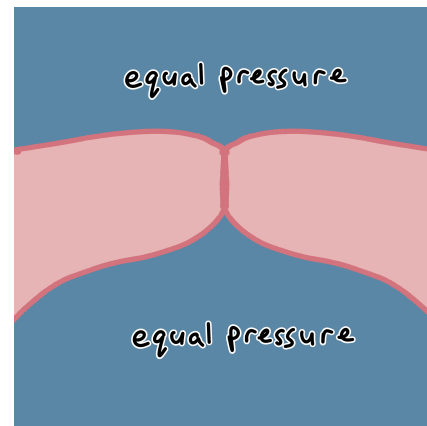
Voicing requires a pressure differential.



voicing                      possible

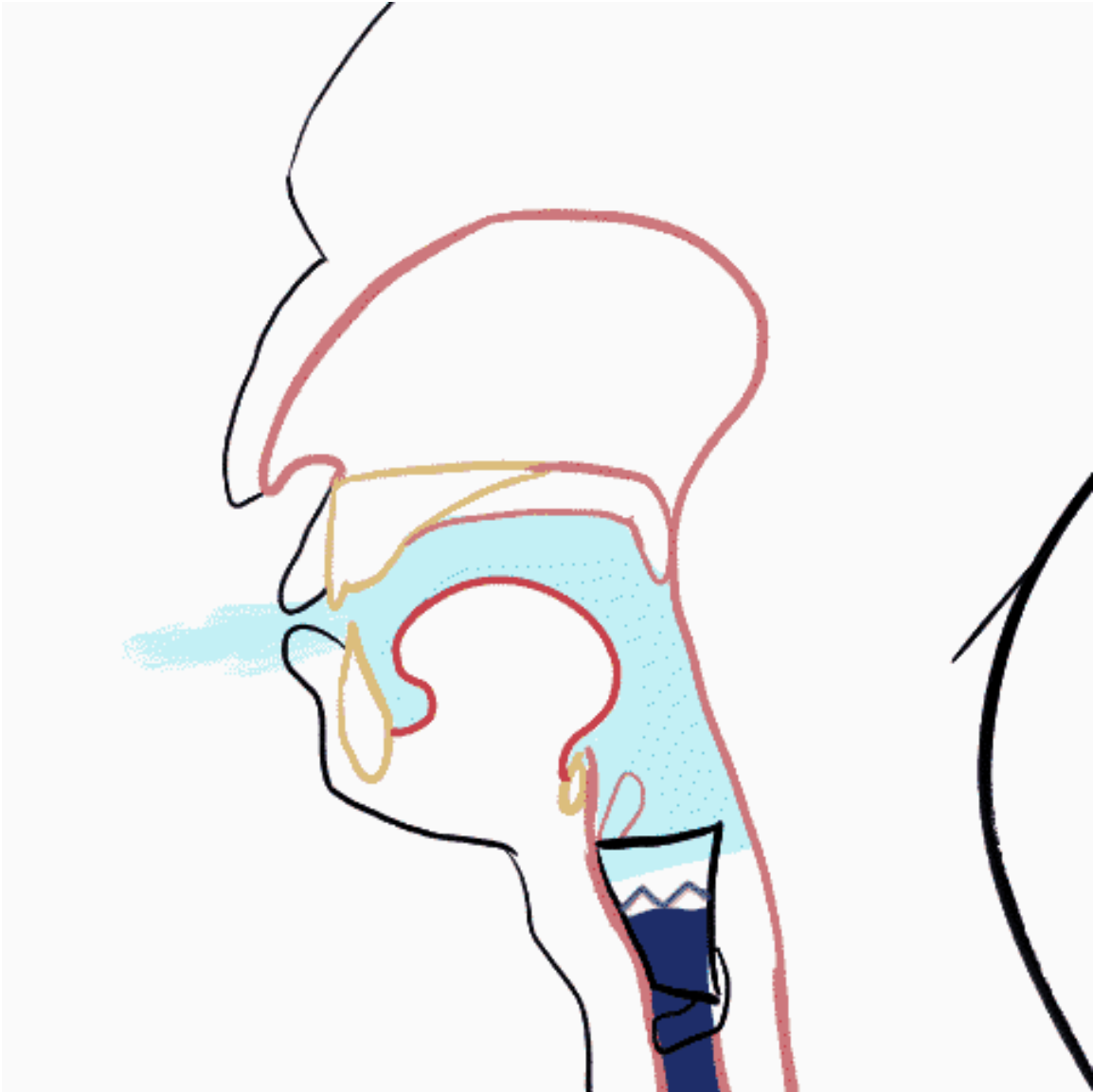


voicing              not              possible

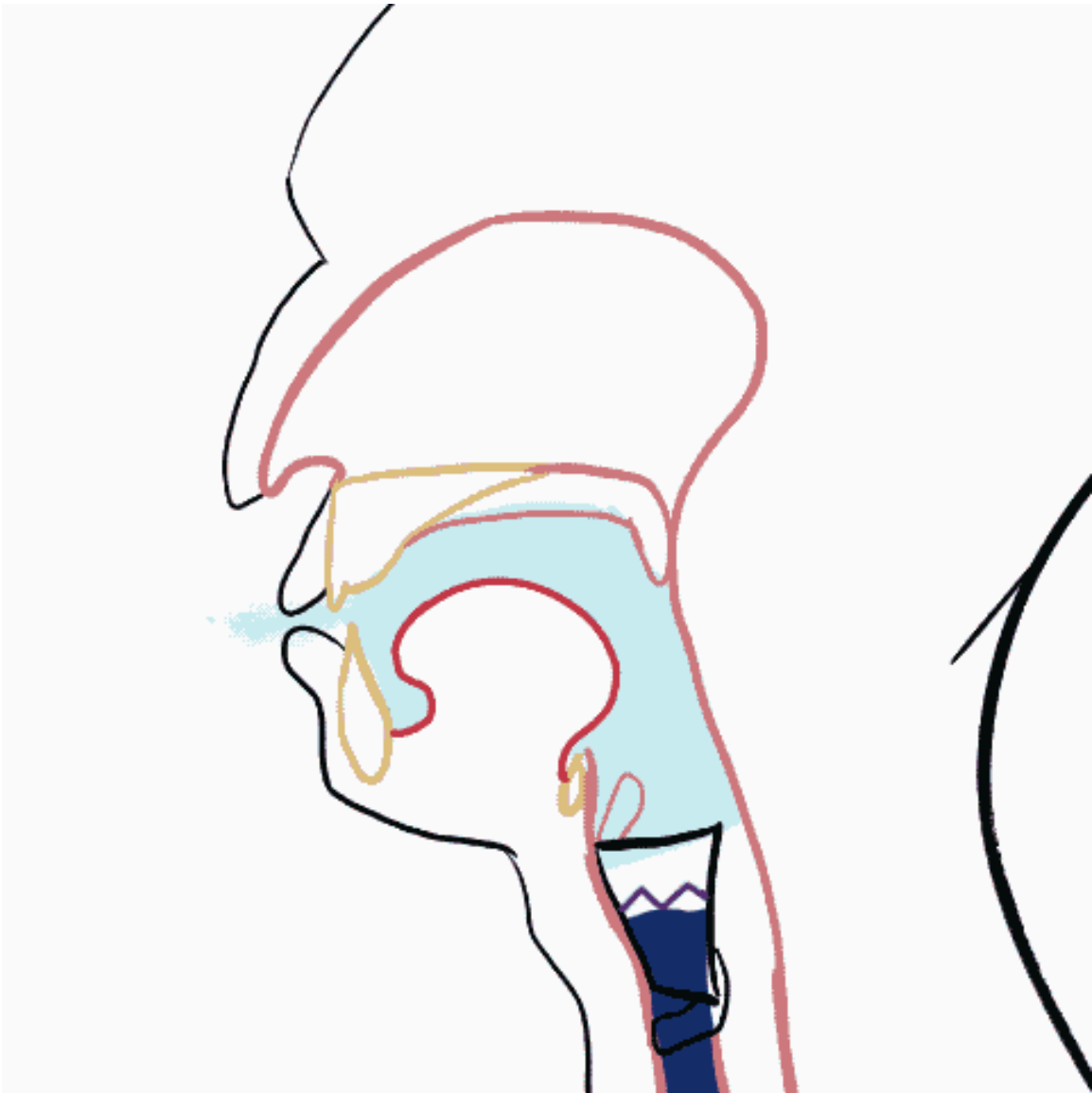


Voicing

No problem for sounds with incomplete closure.



Voicing  
With complete closure, air pressure equalizes.



## Voice Onset Time

### Pulmonic Stops

Pulmonic

Egressive

Stop

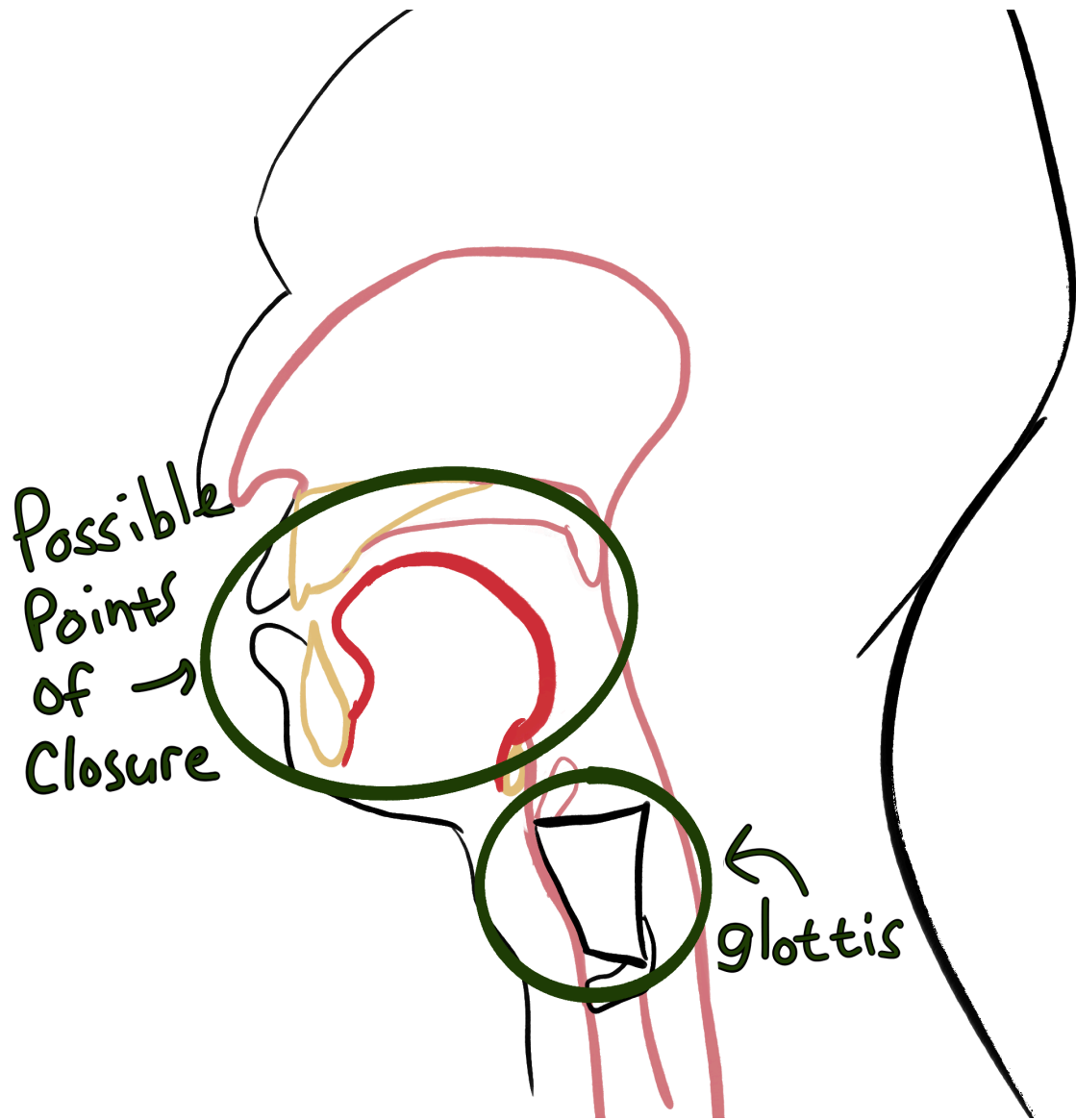
“Air pressure from lungs”

“outward airflow”

“full closure”



## Pulmonic Stops



## Pulmonic Stops

The articulators involved with making stops don't *spatially* interfere with the glottis.

How might constrictions and the glottis interact *temporally*?

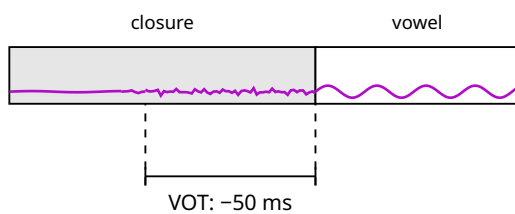


## Fully Voiced Stops



## Fully Voiced Stops

Also known as “negative voice onset time”



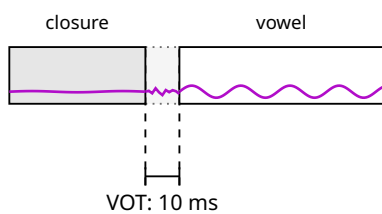


## Short Lag Voice Onset Time



## Short Lag Voice Onset Time

There is a brief lag between releasing the closure and the onset of voicing.



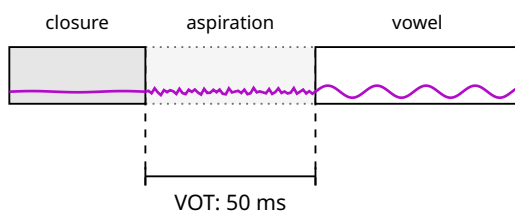


## Long Lag Voice Onset Time



## Short Lag Voice Onset Time

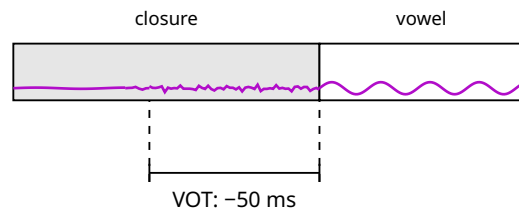
There is a long lag between releasing the closure and the onset of voicing.



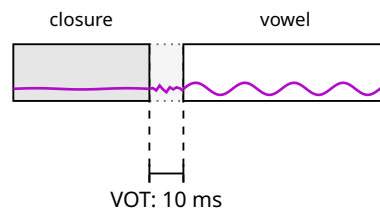


## Voice Onset Time

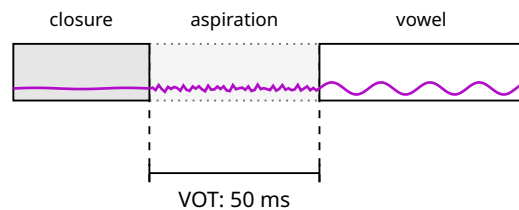
Fully voiced stops



Short lag (a.k.a. Unaspirated Stops)



Long lag (a.k.a. Aspirated Stops)



## Voice Onset Time

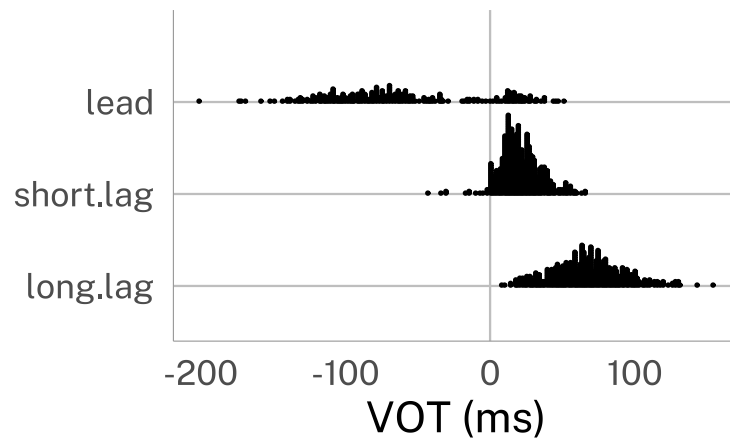
```
vot |>
  filter(
    #poa.broad == "coronal"
  ) |>
  mutate(
    vot.category = fct_reorder(vot.category, desc(vot))
  ) |>
  ggplot(
    aes(vot, vot.category)
  ) +
  geom_vline(
    xintercept = 0,
    linewidth = 0.4,
    color = "grey"
  ) +
  stat_dots(
    color = "black",
```



```

    fill = "black"
  ) +
  labs(
    y = NULL,
    x = "VOT (ms)"
  ) +
  theme_sub_panel(
    grid.major.y = element_line(
      linewidth = 0.4,
      color = "grey"
    )
  )
)

```



Chodroff et al. (2019)

Transcribing different VOTs

| VOT          | IPA               |
|--------------|-------------------|
| fully voiced | [b]               |
| short lag    | ?                 |
| long lag     | [p <sup>h</sup> ] |

Transcribing VOTs

|                 | fully voiced | short lag | long lag                                        |
|-----------------|--------------|-----------|-------------------------------------------------|
| English, German |              | [b d g]   | [p <sup>h</sup> t <sup>h</sup> k <sup>h</sup> ] |
| Cantonese       |              | [p t k]   | [p <sup>h</sup> t <sup>h</sup> k <sup>h</sup> ] |



|                           | fully voiced | short lag | long lag                                        |
|---------------------------|--------------|-----------|-------------------------------------------------|
| Spanish, Dutch, Hungarian | [b d g]      | [p t k]   |                                                 |
| Eastern Armenian, Thai    | [b d g]      | [p t k]   | [p <sup>h</sup> t <sup>h</sup> k <sup>h</sup> ] |

Lisker & Abramson (1964)

Transcribing VOT

Lesson: A lot of transcription is analysis in disguise!

## Non-Pulmonic Stops

### Non-Pulmonic Stops

|                                    | egressive (+ air pressure) | ingressive (-air pressure) |
|------------------------------------|----------------------------|----------------------------|
| glottalic (closure at the glottis) | “ejectives”                | “implosives”               |
| velaric (closure at the velum)     |                            | “clicks”                   |

### Ejectives

1. An oral closure is made.
2. The glottis is tightly closed.
3. The larynx is raised.
4. Pressure builds behind the oral closure.
5. The oral closure is released.
6. The built up pressure rushes out.



## Ejectives



## Ejectives

You can have an ejective at any point of closure, and it's transcribed with ['] (a regular apostrophe)

| bilabial | alveolar | retroflex | palatal | velar | uvular |
|----------|----------|-----------|---------|-------|--------|
| [p']     | [t']     | [ɖ']      | [c']    | [k']  | [q']   |

Ejective fricatives and affricates are also possible.



## Practicing Ejectives

Try holding your breath (thus closing your glottis) and articulating “t t t t” repeatedly.

## Implosives

1. An oral closure is made.
2. The glottis is tightly closed.
3. The larynx lowers.
4. Pressure drops behind the oral closure.
5. The oral closure is released.
6. Air rushes in, normalizing the low pressure.

## Implosives





## Implosives

Implosives can be made at any point of closure, and their IPA symbols tend to have a little hooktop.

| bilabial | alveolar | velar | uvular |
|----------|----------|-------|--------|
| [ɓ]      | [ɗ]      | [ɠ]   | [ʛ]    |

## Clicks

1. A closure is made somewhere in front of the velum.
2. While that closure is maintained, the tongue body moves back.
3. A vacuum is created between the point of closure and the velum.
4. The closure is released.
5. Air rushes in normalizing air pressure.



## Clicks



## Clicks

You can only have clicks with active articulators that don't interfere with the tongue body.

| bilabial | dental | palatal | lateral | post-alveolar |
|----------|--------|---------|---------|---------------|
| [ɔ]      | [ɪ]    | [ʈ]     | [ɬ]     | [ʈʰ]          |

## Clicks In Xhosa [llo:sa]

<https://www.youtube.com/watch?v=n-4U2hfMpnk>



## Bibliography

- Chodroff, E., Golden, A., & Wilson, C. (2019). Covariation of stop voice onset time across languages: Evidence for a universal constraint on phonetic realization. *The Journal of the Acoustical Society of America*, 145(1), EL109–EL115. <https://doi.org/10.1121/1.5088035>
- Lisker, L., & Abramson, A. S. (1964). A Cross-Language Study of Voicing in Initial Stops: Acoustical Measurements. *WORD*, 20(3), 384–422. <https://doi.org/10.1080/00437956.1964.11659830>